

Microproject Proposal

Course Enrollment System in Java

1.0 Brief Introduction

A Course Enrollment System is a console-based application developed in Java that allows a student to register their details and enroll in courses from a predefined list of 25 available courses. The system is entirely console-driven, using standard input and output — no graphics or GUI are involved. The student enters their ID and name at the start, after which they are presented with a menu-driven interface offering three options: enroll in a course, view enrolled courses, and exit. The available courses span a wide range of technology subjects including Java, Python, Data Structures, Artificial Intelligence, Machine Learning, Cyber Security, and many more. The system validates user input and handles invalid entries gracefully, ensuring a smooth and interactive experience throughout the session.

2.0 Aim of Project

The aim of this project is to design and implement a fully functional, console-based Course Enrollment System using Core Java concepts such as classes, objects, methods, ArrayLists, and Scanner for user input. The project demonstrates how object-oriented programming can be applied to model real-world entities — namely Students and Courses — and manage the relationship between them through enrollment operations. It helps reinforce programming concepts including conditional logic, loops, instance variables, and method calls in a practical context. The system is designed to be simple enough for beginners to understand while illustrating important software design principles such as modularity, separation of concerns, and clean class design. Additionally, this project serves as a foundation for understanding how larger academic management systems and database-driven enrollment platforms are structured at a logical level.

3.0 Action Plan :

Sr. No	Details of activity	Planned commencement date	Planned Finish date	Roles and Responsible assigned to whom
1.	Collect data from different Sources (books/internet/etc.)	Week 1	Week 1	All Team Members
2.	Collect data from similar Systems	Week 1	Week 2	All Team Members
3.	Analysis of the collected data & to generate useful information from it	Week 2	Week 2	All Team Members
4.	Prepare required drawings and detailed plan for execution of the work.	Week 2	Week 3	Team Leader
5.	Present generated information visually in the form of appropriate coding	Week 3	Week 4	Developer / Programmer
6.	Use of Equipment	Week 3	Week 4	All Team Members
7.	Prepare the Micro-project report	Week 4	Week 5	All Team Members
8.	Prepare presentation	Week 5	Week 5	All Team Members

4.0 Resources Required

Sr. No.	Name of Resource	Specification	Qty /Units	Remark
1	Computer / Laptop	Windows 10/11, min 4GB RAM	1 per student	For coding & testing
2	JDK (Java Development Kit)	Java SE 8 or above	1 (Free)	Required to compile and run Java
3	IDE / Text Editor	VS Code / Notepad++ / Eclipse	1 (Free)	For writing Java code
4	Command Prompt / Terminal	Built-in OS terminal	1	To run the console game

5.0 References -

Book Name	Author
Java: The Complete Reference	Herbert Schildt
Head First Java	Kathy Sierra & Bert Bates

6.0 Web references: -

<https://www.w3schools.com/java/>

<https://docs.oracle.com/javase/tutorial/>

<https://www.tutorialspoint.com/java/index.htm>